

youlinked from <https://github.com/Nicicalu/OST-CN2-Spik>

OSPF (IGP)
ECMP load balancing, fast convergence, widely used. IP Port 89 on L3 (no TCP, has own header). Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR).
DR Election: Highest IP prio → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same rules, second best candidate
Virtual links: Cannot go through more than one area, can only run through standard non-backbone areas. (eg. not over stubby areas)

ROUTER OPERATION
– Establish neighbor adjacencies and exchange LSAs
– Build the Link-State Database (LSDB)
– Run Dijkstra's SPF algorithm:
Intra-area change: SPF recalculation
Inter-area change: no SPF needed – ABR handles updates
– Build the routing table from SPF results

ROUTER TYPES
O: Intra-area: Local area → Local area. Type 1 and 2 LSAs
O: Intra-area: Local area → Neighboring area. Via LSA 3 through backbone
O E1 or E2: External: Local area → External (eg. BGP), from ASBR
O: L3: Cost = internal + external metrics
O E2: Cost = external metrics. Default external route for OSPF
Cost calculation: Reference Bandwidth(def=100 Mbps)/Interface Bandwidth

AREAS
AS split into **areas** with 32-bit Area ID (eg. 0.0.0.0 = Area 0)
Backbone Area (Area 0): Core of the OSPF domain, must connect to all other areas (directly/virtual links), must be contiguous (no disjointed segments), should not contain end-user networks, summarizes topology of other areas
Non-Backbone Areas: Connect end users and local resources, All inter-area traffic must transit the backbone

ROUTER TYPES
Area Border Router (ABR): Connects two or more OSPF areas, must have 1 interface in backbone, 1 OSPF DB per Area
Internal Router (IR): All interfaces belong to the same area (non-backbone)
Backbone Router (BR): At least one interface in Area 0, Includes ABRs and routers internal to the backbone

AS Boundary Router (ASBR): Connects to external AS/Network (eg., BGP), Advertises external routes into OSPF, can be in backbone or non-backbone

DESIGN RULES
Rule 1: Backbone (Area 0) must be contiguous – no partitions allowed
Rule 2: Every non-backbone area must connect to Area 0

LINK STATE ADVERTISEMENT (LSA) TYPES
(1) **Router-LSA:** All routers list directly connected links (so all outgoing interfaces with state and cost) (**intra-area**)
(2) **Network-LSA:** DR lists all routers and DR in broadcast/multi-access network (**intra-area**)
(3) **Summary-LSA:** ABR advertises networks from other areas, flooded in all areas (and that are not "totally stubby" (**inter-area**)
(4) **ASBR-Summary:** ABR advertises path to ASBRs. ASBR sends Type 1 + external bit, ABR converts to Type 4 and floods (**inter-area**)
(5) **AS-External-LSA:** ASBR/ABR advertises external routes (eg., BGP), flooded to all non-stub areas → only normal areas (**inter-area**)
(7) **NSA External:** Like Type 5 but inside NSA. **Converted to Type 7 by ABR.** Shown in routing table as (O N1 or O N2)

AREA TYPES (ALLOWED LSA TYPES)
Stubby = Blocks external LSA [No 5], Totally = Blocks inter-area LSA [No 3/4]
Default routers (eg. 0.0.0.0) from ABR [DR ABR], Cannot contain ASBR [No ASBR]
Standard (1,2,3,5): Normal
Stub Area (1,2,3): [No 4/5] [DR ABR] [No ASBR]
Totally Stubby Area (1,2): [No 3/4/5] [DR ABR] [No ASBR]
Not-So-Stubby Area (1,2,3,7): [No 4/5], no DR, allows 1 ASBR [LSA 7 → 5]
Totally NSA (1,2,7): [No 3/4/5] [DR ABR] [LSA 7 → 5]

PACKET TYPES
(1) **Hello:** Usually Multicast. Discover, maintain, and verify neighbors. Forms adjacencies. Election of DR/B in broadcast networks. Contains network mask (sending routers' IP), Hello interval (p2p broadcast: def=10s), Options, Priority (election), Router Dead Interval (def=40s), DR/BDR IP. Neighbors.
(2) **Database Description (DBD/DD):** Unicast. Exchange summaries of LSAs during adjacency formation (headers only). Contains Ifr. Max. MTU, Options, I/M/M/S bits (Initial, More, Master-slave bit), DD Seq. Num., LSA Header
(3) **Link State Request (LSR):** Unicast. Request specific LSAs. Contains Link State Type (router/net), Link State ID, Advertising Router (sender address)
(4) **Link State Update (LSU):** Multi/Unicast. **Implicit ack.** Floods newly updated LSAs. Answer to LSR (unicast). Contains Nr. of LSAs, full LSAs information
(5) **Link State Acknowledgment (LSAck):** Multi/Unicast. **Explicit ack.** Confirms receipt of LSAs for reliable flooding. Acks may be grouped together.

SUB-PROTOCOLS
Hello Protocol: Neighbor discovery, parameter negotiation. Maintain logical adjacencies on P2P, P2MP, virtual links. Elect DR/BDR on broadcast and NBMA networks. Continuously send hello packets to maintain bidirectional connectivity, failure to receive = neighbor down (RouterDeadInterval)
Database Sync Protocol: Sync LSDB using DBD packets.

NEIGHBOR STATES
Down: Initial state, no Hello packets exchanged or after RouterDeadInterval interval. Received Hello packet from neighbor but without its own Router ID
2-Way: Seen its own Router ID. DR/BDR election is taking place in NBMA
ExStart: (DBD) Bi-dir comm; highest Router ID=master. Determine init Seq Nr
Exchange: Exchange of DBD packets (LSA headers)
Loading: Missing LSAs are requested. (LSR/LSU)
Full: Databases fully synchronized. Normal operating state

OSPF ROUTING AND ECMP
– Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535)
– Prefer more specific match (CIDR), else O > O > I > E1 > N1 > E2 > N2
– Routes added to RIB/FIB based on computed next hops
– ECMP: Modified Dijkstra supports Equal-Cost Multi-Path if multiple paths have same cost → routes added with multiple next-hops for load balancing

POSSIBLE FAILURES
– No OSPF enabled / Interface not in OSPF / Duplicate router ID
– Passive interface that has to be removed / Virtual Link badly configured
– Area ID or Area Type mismatch on neighbors
– Hello interval or Router dead interval mismatch

IS-IS (IGP)
Widely used by ISPs, Scalable, Fast failure detection & convergence, ECMP, Extendable, Less CPU intensive
ES: End-host devices are called End Systems (ES)
IS: Routers are called Intermediate Systems (IS)

CONNECTIONLESS NETWORK SERVICES (CLNS)
CLNS: ISO Layer 3 datagram service; supports CLNP, ES-IS, IS-IS.
CLNP: Connectionless Network Protocol, similar to IP, used in ISO stack
ES-IS: host → router discovery protocol
IS-IS: router → router. Link-state routing protocol
Integrated IS-IS: Allows IS-IS running with IS-IS
NETWORK SERVICE ACCESS POINT (NSAP)
Variable-length (up to 20 bytes), Identifies 49.0011.0000.0000.0003.00 node, not IF. Example from IP: 192.168.1.24 Area ID (T1) System ID (R3) NSEL → 49.0001.1921.4808.1024.00

Network Entity Type (NET): Unique router identifier. NSEL=0, Starts with 49
Authority & Format Identifier (AFI): NSAP format authority that assigns it
Initial Domain Identifier (IDI): Variable len, identifies administrative domain
Domain Specific Part Format Identifier (DFI): Specifies format
Domain Specific Part (DSP): Var len, contains hierarchical structure of addr
NSEL (N-Selector): always 00 for routers (1b)
System ID: Unique per router (often based on lo-IP/MAC) (8b)
Area ID: Used for routing hierarchy (like OSPF areas) (AFI+IDI+DSP)

PACKET TYPES (ALL IN L1 OR L2)
All PDUs contain Header with shared common fields + specific routing-related information (Type, Length and Value (TLV)) used to extend protocol
IIH (Hello): Build, maintain adjacencies. Includes system ID, holding time, prio
Functions: discover, build, maintain
Interval: 10s (default); DIS sends every 3.3s on LANs
CSNP: Received Hello limit → Holdtime = Interval * Multiplier(def=3)
Multiast: [L1 01-88-02-08-09-14] [L2 01-88-02-08-09-15]
LSN (Link State PDU): Contains topology info including prefixes with costs; unicast on p2p or flooded throughout the area (OSPF: LSA Type 1)
CSNP (Complete SNP): Sent by DIS; lists all known LSAs (OSPF: DBD)
PSNP (Partial SNP): Used to request missing or acknowledge received LSAs

ROUTER TYPES
L1 Router: Within one area, all have same LSDB. No inter-area routing
L2 Router: Backbone router, all have same LSDB. Routes between areas
L1/L2 Router: Acts as both; separates LSDBs, redistributes between levels
ADJACENCIES
Backbone must be contiguous: (L2 connection) L1 ↔ L1-L2 → L2. **NOTE:** L1/L2 routers form both L1 and L2 adjacencies if they are both in the same area
L1: Area addresses match unless configured otherwise
L2: Alongside L1, unless router is L1 only, areas must not match

POINT-TO-POINT (NO DIS)
Initialized by receipt of ISHs, exchange p2p IIHs padded to MTU
CSNP: Sent once at adjacency startup
LSN: Advertises topology changes (link-state info)
PSNP: Acknowledges received LSAs or requests missing ones

BROADCAST/MULTI-ACCESS (DIS REQUIRED)
Not triggered by ISHs, instead broadcast IIHs with all neighbors MAC's
Designated Intermediate System (DIS): Create, update pseudonode LSAs.
Flood LSAs. Similar to DR in OSPF. No backup DIS.
DIS Election: Highest priority (0–127) (Cisco def=64) = Highest SNPA (MAC)
Preemption: Enabled → higher prio router automatically takes over DIS Role
Pseudonode: Carried out by DIS. Multiaccess links. Separate for L1/L2
Passive interfaces: Advertise network prefixes without adjacency forming
Default interface metric = 10. Lower Metric better: L1 intra-area > L2 intra-area > Leaked L2 → L1 (internal metric) > L1 external (external metric) > L2 external (external metric) > Leaked L2 → L1 (external metric)

ROUTING
Summarization occurs when routers enter an IS-IS level (L1 → L2, L2 → L1).
LEVEL 1
Intra-area routing only (L1 Area like OSPF Totally Stubby Area)
If routers install a **default route** to nearest L1-L2 for inter-area traffic
L1-L2: Do not advertise L2 routes into L1 area (unless router leaking active)
Set **Attached bit** to signal L2 connectivity to backbone
Distribution Bit: Set to 1 on L2 → L1; blocks re-advertisement L1 → L2.
Route-Leaking: injects a more specific route to L1 to improve routing
LEVEL 2
Routing between areas (**inter-area**). L1-L2 routers inject L1 routes into L2 topology. L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP

IS-IS VS OSPF

Feature	IS-IS	OSPF
Layer	L2 (CLNS)	L3 (IP, Port 89)
Encapsulation	No IP, uses TLVs	IP packets
Hello Type	IIH	Hello packet
Area Model	L1/L2	Backbone + Areas
Metric	Cost (default 10)	Cost (bandwidth)
Router ID	System ID (6B)	32-bit router ID
Adj. Types	L1/L2, L1/L2	DR/BDR, P2P
Scaling	Per level (L1/L2)	Per area
Routing Info	Large-scale ISL core	Enterprise/campus
Type	TLVs (flexible)	Fixed LSA types
Shortest Path calculation	Link-State	Link-State
	Dijkstra	Dijkstra

POSSIBLE FAILURES
– Wrong addressing (NET, Sys-ID, Area ID) / Interface type (P2P, broadcast)
– Wrongly config LSP, L1/L1, L1/L2 routers (eg. no L1-L2 as transit)
– Authentication mismatch (TLV auth → configure identical IS-IS auth)

BGP (EGP)
Path-vector routing protocol (doesn't contain full topology). Stable, flexible, scalable. Based on TCP. Loop prevention using AS_PATH

IGBP
Routers in same AS, AD=200, more trusted (lower security overhead). AS-Path and Next-hop not modified

SCALABILITY
– iBGP does not re-advertise routes between iBGP peers → full mesh required (caused no loop prevention)
– Session count = $n \times (n-1) / 2$ (e.g., 5 routers = 10 sessions, 10 = 45)

ROUTE REFLECTORS (RR)
Solves iBGP full-mesh (**split horizon**) scaling by selective route refl. Clients only peer with RR; unaware they're clients. Only RR needs specific config
From non-client: RR advertises to clients only
From client: RR advertises to all clients + non-clients
From eBGP peer: RR advertises to all clients + non-clients

EBGP
Routers in different ASes, AD=20, stricter policy enforcement. Prepends ASN to AS-Path, modifies next-hop IP. Fails if own ASN in AS-Path. TTL=1
AS-Override: Allows reuse of same ASN across different customer sites (eg., Swisscom); rewrites ASN to avoid loop detection
CONFIG
next-hop-self fixing iBGP: neighbor (neighbor IP) next-hop-self (overriding)
AUTONOMOUS SYSTEM NUMBERS (ASN)
– Unique ID for each AS; required for Internet routing; no 294
– Private: [Legacy 16-bit 64512-65535] [32-bit 4200000000-4294967294]
PEERING / NEIGHBORS
– Two routers with a BGP TCP session (port 179) are called peers/neighbors
– Each BGP router is a **BGP speaker**
– Supports CIDR, route aggregation, decisions based on policies/rules

PATH ATTRIBUTES
Used for route control and policy enforcement
Well-known mandatory: Always present (eg., AS-Path, Origin, Next Hop)
Well-known discretionary: Optional but recognized by all (eg., Local Pref)
Optional transitive: Passed between ASes (eg., Community, Aggregator)
Optional non-transitive: Not passed across ASes (eg., MED, Weight)
NLRI: Routing table info: prefix, prefix length, and associated path attributes

MESSAGES
OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params. **Hold Time:** Heartbeat in seconds (Cisco def=180s), reset by KEEPALIVE/UPDATE: 0 = session down. **BGP Identifier:** 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention
KEEPALIVE: Sent every 1/3 of Hold Time; ensures neighbor liveness
UPDATE: Advertises new prefixes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE
NOTIFICATION: Sent on session error (eg. hold timer expired); terminates session immediately

NETWORK STATEMENTS
Advertise a specific network prefix to BGP peers (only the best is advertised, even if multiple exist). Must exist in the RIB so that it's imported in BGP table
Connected: next-hop: 0.0.0.0, origin attribute: 1, weight: 32768 (Cisco)
Static/RP: next-hop: NH in RIB, MED: IGP metric, rest same as above

BEST PATH CALCULATION
– BGP maintains all received paths per prefix but advertises only the best one
– Best path is installed in RIB; recalculated on:
– Next-hop reachability change
– Interface failure to eBGP peer
– Redistribution change
– New/withdrawn path received
– Influence:
– Outbound BGP policy → inbound traffic behavior
– Inbound BGP policy → outbound traffic behavior

BEST PATH SELECTION (IF PREFIXES ARE THE SAME):
1) Prefer highest **Weight (Cisco-specific)**, local to router)
2) Prefer highest **Local Preference** (global within AS)
3) Prefer routes **originated by the router** (only small i path, NH 0.0.0.0)
4) Prefer **shorter AS path** (only length is compared)
5) Prefer **lowest origin type** (IGP < EGP < Incomplete [1 on origin])
6) Prefer **lowest MED** (Multi-Exit Discriminator) (also called metric)
7) Prefer **external (EBGP) over internal (IBGP)**
8) For iBGP: prefer path with **lowest IGP metric** to next-hop
9) For eBGP: Prefer **oldest** (more stable) path
10) Prefer **lowest BGP router ID**
11) Prefer path from **lowest neighbor IP address**

ROUTE FILTERING
– Filters control which routes are received/advertised
– Used for security, traffic shaping, memory optimization
– Tools: **prefix-list** (IP), **filter-list** (AS-path), **route-map** (flexible match/set)

COMMUNITIES
32b optional, transitive tag (eg. ASn: value, 65000-100). Used to mark routes for policy control across ASes. Can be added, modified, or removed at each hop

ENTERPRISE CONNECTIVITY OPTIONS
SingleDual: denotes how many links there are
Multi-Homed-Homed: denotes how many ISPs are connected
Single-Homed: One ISP, one link (BGP or static), simple but no redundancy
Dual-Homed: One ISP, two links (routers), Redundancy within same provider
Multihomed: Multiple ISPs, improved redundancy and routing control, avoid being a transit – advertise only customer-owned prefixes
Dual-Multihomed: Multihomed, but two links per ISP. Most redundancy

TRAFFIC ENGINEERING (TE) - INBOUND VS OUTBOUND
Local Pref: [Out] [highest T] Lower local pref on backup path
MED [in] [lowest T] Higher MED on backup path
AS-Path Prepending: Add multiple own ASN on backup path
TE Limitation: AS controls [Out], [in] control limited - ISPs may ignore
TE with Aggregate: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for failover
Transit: One ISP provides reachability to all destinations
Peering: ISPs provide each other reachability to portions of their routing table

TRANSIT VS. PEERING
Transit: ISP provides full reachability (paid relationship)
Peering: ISP exchange selected routes; equal relationship, usually unpaid
AS Path Filtering: to avoid getting transit; ip as-path access-list 10 permit ^ \$

INTERNET EXCHANGE POINT (IXP)
Facility where networks exchange traffic via BGP peering, **routing policies** determine which routes are advertised/accepted. Reduces transit costs, latency, and offloads upstream links

PUBLIC PEERING
Members peer via shared switch fabric and a **route server**, which distributes routes but stays out of data path (NEXT_HOP unchanged). Simplified setup (one legal contract), minimal policy control; one BGP session to route server

PRIVATE PEERING
Direct BGP sessions between two parties (1:1), requires one session and legal contract per peer. May use public or private interconnects. Full policy control per neighbor. **Examples:** Equinix, SwissIX (non-profit)

RESOURCE PUBLIC KEY INFRASTRUCTURE (RPKI)
– Framework to validate that prefix is legitimately originated by a specific ASN
– **AS Path Hijacking:** one hijacker and one identical mis-originations (route leaks)
– RPKI validates origin only – not the AS-path

KEY COMPONENTS
Trust Anchors (TAs): Root CAs of the 5 RIRs; issue certs for resource holders
Route Origin Authority (ROA): Digitally signed object, authorizes ASN to announce prefix (AS, prefix, that AS can validate, max prefix length)
RPKI Validators: downloads+verifies+stores Validated ROA Payloads (VRPs)

RPKI VALIDATION STATES
Valid: Prefix + ASN match ROA
Invalid: Prefix found, but ASN mismatch or prefix too long
Not Found / Unknown: No matching ROA

VALIDATOR DETAILS
– Use TALS (Trust Anchor Locations) to fetch data from RIRs
– Validate cryptographic signatures (via x.509 certs with RFC 3779)
– Outputs VRFs; invalid objects are discarded
– Update frequency: >=24h (recommended), ~30–60min (practice)
– RRDP (RFC 8182) replacing rsync (uses HTTPS)

MONITORING
Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization

MULTICAST
Use-cases 1-to-Many: Streaming, software updates, music-on-hold
Use-cases Many-to-Many: Gaming, VR, stock data, group chat
Benefits: Efficient bandwidth, lower server/CPU load, no redundancy
Properties: UDP-based. Apps must handle drops, duplicates, out-of-order
Source: Sends to group IP; doesn't need to join
Receiver: Must explicitly join group to receive traffic

MULTICAST ADDRESS RANGES
224.0.0.0 – 224.0.0.255: Link-local, TTL=1 (not forwarded by routers) (IGMP)
224.0.1.0 – 224.0.1.255: Reserved by IANA, routable (NTP)
232.0.0.0 – 232.255.255.255: Source-Specific Multicast (SSM)
239.0.0.0 – 239.255.255.255: Globally scoped multicast (routeable)
239.0.0.0 – 239.255.255.255: Administratively scoped (private space)

BROADCAST BASICS
L3 routes between subnets; L2 floods within same subnet
L2 (Bridging): MAC xxxx.ffff.ffff switches flood to all ports in VLAN
L3 (Routing): 255.255.255.255: local broadcast, not routed. Directed broadcast for specific subnet (eg. 10.1.1.255) can be routed if enabled

L2: MULTICAST & MAC MAPPING
1) **Get IP:** Example multicast IP address: 239.5.5.5
2) **Convert to Binary:** 239.5.5.5 = 11011111.00000101.00000101.00000101
– Take only the last 23 bits: 00000101.00000101.00000101
3) **Map to MAC Prefix:** Fixed 0100 5E – Map last 23 bits to: 0100 5E05.0505

INTERNET GROUP MANAGEMENT PROTOCOL (IGMP)
Purpose: Manages group membership for IPv4 multicasts on each segment
IGMPv1: Basic join via query-response mechanism, no explicit group leave. Router sends **membership queries** every 60s to 224.0.0.1, removes group after timeout if no report received. Receiver doesn't know multicast source
IGMPv2: Adds **Leave Group** message. Supports **Group-Specific Queries** (eg. when someone leaves group (anyone still interested in group xy?), reducing broadcast overhead. Receiver still doesn't know who the source is)
IGMPv3: Adds **source filtering** (include/exclude lists). Enables **Source-Specific Multicast (SSM)** – receiver requests traffic only from selected source(s), **no need for Rendezvous Point (RP)** anymore. Adds support for application-level access control and filtering. Can also be used in ASM (Any Source Multicast), but mainly with SSM. (224.0.0.22 all IGMPv3 routers)

IGMP SNOOPING
– Without snooping: multicast = broadcast on VLAN
– With snooping: switch listens to IGMP messages & builds a forwarding table
Default: snooping is enabled; switch needs IGMP Querier to operate

REVERSE PATH FORWARDING (RPF)
To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of

RENDEZVOUS POINT (RP)
Used in Shared Tree (*,6) setups with PIM-SM, acts as common meeting point. **RPF check:** incoming traffic toward the RP (not the source). Once source is known, routers may switch to a Source Tree (S,6). Triggered by SPT join timer expiration/RPF check confirming better route/data rate exceeding

SHARED TREE VS. SOURCE-BASED TREE
Source-Based Tree (S,6): Built and added to **route table** when router receives an (*,6) join/report from IGMP host. s = known source; g = group
Shared Tree (*,6): IGMP host sends **membership report** (IGMP Join). Router adds (*,6) entry to **multicast routing table** = any/unknown

PROTOCOL INDEPENDENT MULTICAST (PIM)
Only between routers. Relies on unicast routing table (RIB) for multicast forwarding decisions and RPF. Protocol-independent.

PIM DENSE MODE (PIM-DM) - NO RP
Push Model: Floods multicasts to all interfaces; prunes where no receivers
(1) **Flooding:** Source sends traffic to multicast-enabled links using unicast RIB
(2) **Distribution Tree:** Initially entire network (shared tree rooted at source)
(3) **Prune Messages:** Routers without interested receivers send prunes up stream to remove themselves from the tree
(4) **State Maintenance:** Routers track source, receivers, interfaces to forward/prune per group

PIM SPARSE MODE (PIM-SM) – ASM, SSM
Push Model: Multicast traffic only sent where requested (**explicit**) joins. Works with IGMP to detect interested receivers, uses unicast routing for forwarding
(1) **Join/Prune:** Routers send explicit join/prune messages to request or stop receiving multicast for group (g) to other routers
(2) **Forwarding:** Routers only forward multicast packets for group (g) on interfaces from which explicit joins were received

ANY SOURCE MULTICAST (ASM)
– Works with **IGMPv1 or IGMPv2** → receiver does not know the source
– Receiver sends **IGMP Join (*,g)** to its first-hop router
– First-hop router forwards **PIM Join (*,g)** hop-by-hop toward the RP
– Sources send multicast traffic to the RP via a **PIM Register** tunnel
– All routers in the multicast domain must know the RP location

SOURCE SPECIFIC MULTICAST (SSM)
– Receiver subscribes using **IGMPv3**, providing both source (s) and group (g) to the first-hop router (**explicit**) join
– No RP required → **PIM-SSM builds only (s,g)** Shortest Path Trees (SPT)
– **No shared tree (*,g)** model used: SSM is source-directed
– IANA reserved **232.0.0.0/8** for SSM in IPv4
– Join mgs forwarded hop-by-hop toward source to establish forwarding path
– Uses unicast routing table (RPF) to maintain loop-free delivery

PIM SPARSE-MODE MODE
PIM Sparse-Mode: Multicast traffic is forwarded only on request
PIM Dense-Mode: Push model – traffic is flooded everywhere, then pruned
Sparse-Mode: Supports both per multicast group, depends on RP availability
PIM DENSE MODE VS. PIM SPARSE MODE USE-CASES
DM: Small/lightly scoped networks where most devices need multicast
SM: Large-scale/distributed env where multicast receivers are few/spread out

NETWORK DESIGN
Pillars: Scalability, Speed, Availability, Security, Manageability → overall cost
AVAILABILITY
MTTD: Mean Time To Detect
MTTI: Mean Time To Identify
MTR: Mean Time To Repair
MTRRS: MTT Restore Service
MTBF: Mean Time Between Failures
MTBSI: MTB Service Incidents
MTBF combined: $(\sum_{i=1}^n 1/MTBF_i)^{-1}$
MTBF parallel: $\sum_{i=1}^n 1/MTBF_i$
Availability: $MTBF / (MTBF + MTR)$
Better Av: $MTR \rightarrow 0 \wedge MTBF \rightarrow \infty$
REDUNDANCY
Adds reliability, increases MTBF but increases MTR (complexity)
Backup Paths: Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed
Load Balancing: ECMP, EtherChannel, Port-Channel
TOPOLOGY
Star (flexible → office), Ring (cheap → city), Bus, Tree, Full Mesh
HIERARCHICAL DESIGN
Core: L3, Highspeed backbone, no policy, scalable/redundant, no security
Distribution: L3, policy control, HSRP/VRRP, loop protect, small fault domain
Access: L2, Connect end devices, high port count, port security, QoS marking
Collapsed Core: Combines Core + Distribution (small/medium networks)
FLAT DESIGN
Any-to-any; small networks, MPLS/LAN setups, lacks scalability and control
FABRIC DESIGN (EMERGING TECHNOLOGIES)
Modern design using **Underlay** (transport, eg. IP, MPLS) / **Overlay** (logical virtual topology, eg. EVPN) with **Software-Defined Networking (SDN)** and **Software-Defined Access (SDA)**
ENTERPRISE CAMPUS
– 1000s of users, multiple buildings, one physical location, one company
– Multiple interconnected LANs, connected via Ethernet and Wireless
– Flat design for client access, Hierarchical design for the rest (per building)
FIRST HOPE REDUNDANCY PROTOCOLS (FHRP)
Ensures default gateway is always reachable, PCs can only have one. Subnet must be on a single router
Virtual Router Redundancy Protocol (VRRP): Shared virtual IP, real MACs per router. Master router handles forwarding, rest as backup
Hot Standby Router Protocol (HSRP): (Cisco) Shared virtual IP + virtual MAC. One active, others in "hot standby" → faster
Gateway Load Balancing Protocol (GLBP): (Cisco) HSRP + Load balancing + redundancy. **Active Virtual Gateway (AVG):** Answers ARP and sends virtual MACs of AVFs. **Active Virtual Forwarder (AVF):** Forwards traffic

DATA CENTER

Parameters	Tier 1	Tier 2	Tier 3	Tier 4
Uptime guarantee	99.971%	99.741%	99.982%	99.995%
Downtime per year	<28.8 hours	<22 hours	<1.6 hours	<26.3 minutes
Price	\$	\$	\$\$\$	\$\$\$\$
Compartmentalization	No	No	No	Yes

DESIGNS
Top of Rack (ToR): switches per rack. less cabling, easy expansions/exchanges per rack, scalable glass fiber, ideal for high service density, more switches, more ports, more L2 Srv-2-Srv traffic, more STP
End of Row (EoR): switches per row. Fewer ports, less switches, higher utilization of ports, switches all at one place, better L2 availability between racks, more cabling
TRAFFIC DIRECTIONS
North-South: Between external networks (client-server, in/out DC)
East-West: Within DC (eg., server-server, storage)
THREE-TIER ARCHITECTURE
– Access → Aggregation → Core (like Hierarchical)
– Optimized for North-South traffic, Not ideal for East-West (VMs, containers)
LEAF-SPINE ARCHITECTURE
– Two-tier: Leaf switches (access) connect to Spines (core)
– High performance, low latency, Scalable, ideal for East-West traffic
WAN
Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**
Bandwidth: App needs low latency, reserve for VoIP
Control/Security: Trust provider? No full control
Availability: Redundancy, SLA for failures
Mgmt: Inband vs out-of-band

PRIVATE WAN
Point-to-Point: Leased L2 line (Ethernet); monthly fee; private circuit; fast
Dart Fiber: Physical fiber lease; costly; ISPs prefer selling lambdas
Coarse-grained Division Multiplexing (CDM): x15, cheaper, 120km, active MUX
Fine-grained Division Multiplexing (FDM): mirrors to aggregate separate signals
Dense Wavelength Division Multiplexing (DWDM): x80, 10Gb/s, 1000km, active MUX, more precise, uses software ("smarter"), wideband
Connection-oriented: Predefined path, packet IDs (ATM, Frame Relay)
Connectionless: No setup, full address in each packet (Ethernet, MPLS VPN)
Circuit-Switched: Requires signaling protocol to establish e2e circuit
Packet-Switched: Expects network to know how to forward packets
PUBLIC WAN
Site-to-Site VPN: Fixed locations, devices locked to IP addr
Remote Access VPN: Changing locations, devices not locked to IP addr
Software-Defined WAN (SDWAN): connects LANs across large distances using controlling software that works with a variety of networking hardware.

MULTI PROTOCOL LABEL SWITCHING (MPLS)
– Advantage of eBGP at PE-CE: No mutual redistrib., same routing process
– Control plane (

Label Forwarding IB (LFIB): Built from LIB; used for actual forwarding decisions. (LJFIB only contains currently best LSP (decision: Routing Protocol) **Control Plane:** Builds routing/labeling tables (RIB, LIB) **Data Plane:** Forwards packets (FIB, LFIB); pushes/swaps/pops labels

MPLS HEADERS
4-byte header before IP. With MPLS VPN, there will be a stack of two headers **Label (20b):** actual MPLS label **EXP (3b):** QoS/CoS, Now called Traffic Class (TC) **S-bit (1b):** Bottom of label stack indicator, 1 = True = last label before IP header **TTL (8b):** time-to-live (equal to IP TTL)

TTL
- Ingress PE decrements IP TTL field & copies IP TTL field into new MPLS TTL
- Routers decrements MPLS TTL
- Egress PE decrements MPLS TTL, pops final MPLS header, copies IP TTL
- If provider doesn't want to expose MPLS network: disables **MPLS TTL propagation** on PE, PE sets MPLS TTL=255, egress PE leaves original IP TTL unchanged → network appears as single router hop from TTL perspective

LABEL DISTRIBUTION PROTOCOL (LDP) - CONTROL PLANE
- Distributes labels to neighbors, triggered by new IP route in FIB
- **Hello messages** sent via UDP (Port 646) to 224.0.0.2 to discover neighbors
- TCP (646) connection used to exchange label bindings (prefix to local lib)
- Routers advertise all local bindings after TCP session is up
- Label mapping used to build LIB → LFIB
- LDP router ID must be reachable (via routing table)
- Each router manages local labels independently
- Relies on underlying routing protocol for best route & loop prevention

MPLS L3 DATA PLANE
Outer label: Transport label (LDP); identifies LSP between ingress/egress PE
Inner label: VPN label (MP-BGP); identifies customer VRF
Push: Ingress PE; classify and label packets
Swap: P router; replaces label, forwards based on new label
Pop: Egress PE removes label; sends original packet to CE
Penultimate Hop Popping (PHP): penultimate router removes the outer MPLS label before forwarding to egress PE, default enabled, ISPs disable it

VIRTUAL ROUTING AND FORWARDING (VRF) TABLES
- Virtual router inside a PE. Maintains isolated RIB + FIB and separate routing progress per CE (VPN isolation)
- Exist per MPLS-aware PE router; one per attached customer

ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)
Route Distinguisher (RD): *Uniquely identifies* VPN routes (per VRF) → allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN. Forms **VPNv4 NLRI (MP-BGP): RD:IPv4 prefix** (12 bytes: RD 8b, IP 4b)
Route Target (RT): Controls **route import/export** between VRFs. Uses as **extended BGP community path attributes**. Typically in the form **ASN:nn** or **IP:nn**, e.g., 65000:100, 1:10

How RTs Work: Route is tagged with an RT when advertised by BGP. Other VRFs import the route if the RT matches their import policy. Allows overlapping or shared connectivity between tenants (e.g., shared services).

OVERLAY TECHNOLOGIES
Overlay network (virtual) sits on top of the **underlay network** (physical). **GRE Routing Encapsulation (GRE):** Encapsulates data packets, unencrypted, NP2. Enables usage of unsupported protocols. Carries separate protocol/original packet inside carrier/transport IP header routed by underlay

SEGMENT ROUTING (SR)
Source routing: Sender defines full path using Segment (= Instruction) List
Segment Identifier (SID): Each SID = 1 instruction (e.g., forward via ECMP, specific iface, or to a service)
Segment List: Ordered SID list carried in packet header; defines full route
State in packet: No per-flow state, intermediate nodes follow SID instructions
No new control plane: Uses existing protocols (OSPF, IS-IS, BGP) with extensions; no LDP or RSVP-TE needed
Data plane: Can operate over MPLS or IPv6
Simple but powerful: Enables TE, fast reroute, policy routing

SEGMENT LIST OPERATIONS
Push: Insert SIDs into packet; set active SID (top of list)
Continue: Active SID not yet completed; keep processing it
Next: Current SID completed, activate next SID in list

SEGMENT RANGES
Global Segments: Known and supported by all SR nodes in the domain, installed in forwarding tables across the network (e.g. "Forward packet according to shortest path to Node1"). Defined in the SR Global Block (SRGB) and should be consistent across all nodes
Local Segments: Defined and installed only on originating node, not forwarded by others, but must be understood network-wide (e.g. "Forward packet on interface to Node2"). Defined in the SR Local Block (SRLB) and are specific to the local SR node.

CONTROL PLANE SEGMENT TYPES
IGP Prefix Segment: Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries. "steer traffic along ECMP-aware shortest path"
IGP Next-Hop Segment: Global SID for a specific node (shortest-path forwarding)
IGP Adjacency Segment: Global SID for a group of nodes; traffic sent to nearest IGP Adjacency Segment: Local SID; direct link to neighbor
L2 Adjacency SID: Local SID for Layer-2 segment (e.g., Ethernet link)
Combining Segments: End-to-end paths can mix IGP and BGP segments. Traffic to BGP Anycast → more ECMP in data centers

TECHNIQUES
- Reuses existing MPLS data plane – no hardware change needed
- Segments = MPLS labels; Segment List = label stack (top = active)
- Segments distributed via IGP/BGP; no LDP necessary (but possible)
- Supports both IPv4 and IPv6 networks
- PUSH= PUSH, CONTINUE= SWAP, NEXT= POP

BENEFITS OF SEGMENT ROUTING
Benefits: Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffic eng. (Delay, Bandwidth, Packet Loss, TE metric, Controller, Source-Node), Seamless deployment, Robust
Source Routing: Balances distributed intel with centralized optimization
Traffic Engineering (TE): Optimizes network performance by analyzing and controlling data flow to reduce congestion and improve QoS

DRAWBACKS OF TRADITIONAL NETWORKS
Control Plane: LDP/RSVP-TE adds complexity
Scalability: Per-flow/path limits growth; LSP, signaling overhead increase fast
VIRTUAL EXTENSIBLE LOCAL AREA NETWORK (VXLAN)
Issues of L2: STP. Max amount of VLANs (4094), Large MAC Address tables
VXLAN: Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation
VXLAN Network Identifier (VNI): 24-bit identifier (up to 16 million segments) that defines the VXLAN broadcast domain.

Virtual Tunnel Endpoint (VTEP): Device (switch, router, or host) responsible for encapsulating/de-encapsulating VXLAN traffic.
Network Virtual Interface (NVE): Logical interface on a VTEP used for VXLAN tunnel operations.

FRAME FORMAT
- Original Ethernet frame → VXLAN Header → UDP → Outer IP Header
- The VXLAN header contains the 24-bit VNI and flags.
- Outer headers allow Layer 2 frames to traverse IP underlay networks.

VIRTUAL NETWORK IDENTIFIER (VNI)
- 24-bit VXLAN Network Identifier uniquely defines VXLAN segments.
- Replaces traditional VLAN IDs (12-bit)
- Used by VTEPs to map traffic into corresponding Layer 2 domains.

VXLAN TUNNEL ENDPOINT (VTEP)
Connects the overlay (VXLAN) and underlay (IP) networks. A VTEP can have multiple VNI interfaces, but they associate with the same VTEP IP interface.
Software VTEP: Located on hypervisors using virtual switches.
Hardware VTEP: Located on routers/switches with ASICs for performance.

VTEP IP Interface: Connects to the underlay network, handles encapsulation.
VNI Interface: Virtual interface per segment (like SVI); handles segregation of Layer 2 domains.

MAC ADDRESS LEARNING
Each VTEP maintains a VXLAN mapping table linking destination MAC addresses to remote VTEP IPs.
Data plane: reactive, Flood and learn, MAC learned from incoming traffic, BUM via ingress replication or multicast underlay, No control-plane needed
Control Plane: proactively MP-BGP EVPN/MAC-to-VTEP advertised via BGP
- No flooding needed for known MACs, ARP suppression, silent-host handling
Flood and learn: Host H1 sends ARP request, switches learn H1's MAC → ARP request is flooded to H2 → H2 responds; switches learn H2's MAC.
Static VXLAN: Manual MAC-to-VTEP mappings. Sends copy of BUM traffic to all participating VTEPs. Doesn't scale well; BUM traffic is inefficient.
Multicast VXLAN: VTEPs join multicast groups per VNI. Scales better, offloads BUM replication. 20+ VTEPs = too much traffic, doesn't scale well
MP-BGP EVPN: Modern solution using BGP as control plane. Dynamically learns MAC/IP info.

ETHERNET VIRTUAL PRIVATE NETWORK (EVPN)
Overcome flood-and-learn limitations, doesn't rely on data plane learning, utilizes robust control-plane MP-BGP, works with different encapsulation techniques (VXLAN, MPLS), excellent scalability, L2 and L3 Support.

LAYER 2
L2 bridging across L3 networks
BGP Control Plane: Distributes MAC info (no flooding → efficiency)
VXLAN Overlay: Encapsulates L2 in L3 UDP (data plane)
Multi-Tenancy: via VNI segmentation
Redundancy: All-active multihoming, ECMP, fast convergence

USE CASES
- Multi-tenant datacenter interconnects (DCI)
- Extending L2 over WAN between remote sites
- Scalable, segmented L2 fabrics
AUTODISCOVERY VIA ROUTE REFLECTORS
- RR reflects EVPN routes to other PEs
- RR doesn't participate in EVPN or pseudowires
- RR needs only **address-family l2vpn evpn**
- L2VPN RIB stores endpoint/VNI info for control plane
BGP UPDATE from spines contain **ORIGINATOR_ID** (origin leaf)
HOST DETECTION
- Host connects to VTEP → MAC learned locally
- VTEP advertises MAC + L2VNI via BGP EVPN
- MAC learning follows normal Ethernet semantics

INGRESS REPLICATION (IR)
Forwarding BUM traffic. When Multicast underlay network is not used, IR handles multi-destination traffic (ARP → unicast). Ingress device replicates BUM packets, sends them as separate unicast to egress devices.

EARLY ARP TERMINATION (ARP SUPPRESSION)
- Avoids flooding ARP requests
- VTEP queries control plane for MAC/IP/VNI mapping
- If known → direct unicast (no broadcast)
Silent Host Flow (Fallback)
- If IP/MAC unknown → ARP sent via **ingress replication**
- Replicated ARP request goes to remote VTEPs
- Remote VTEPs respond → update reflected to all VTEPs
- Future traffic uses updated BGP mapping

LAYER 3
By adding L3 features, data routing efficiency + smooth connections improve
EVPN ROUTE TYPES
(1) Ethernet Auto-Discovery Route (3) Inclusive Multicast Route (4) Ethernet Segment Route (6) Selective Multicast Ethernet Tag Route (7) IGMP Join Synchrony Route (8) IGMP Leave Synchrony Route
Type 2 - Host Advertisement: Advertises host MAC (mandatory), optionally IP, along with L2VNI and optionally L3VNI. Used for MAC learning, ARP suppression, and host mobility. Sent when host connects to VTEP
Type 5 - Subnet Advertisement: Advertises IP prefix + prefix length with L3VNI. Used for inter-subnet routing. VTEP redistributes connected/static/dynamic IP routes. Additional attributes: L3VNI, extended communities.

HOST DETECTION
- Host sends ARP/ND to local VTEP
- VTEP learns MAC/L2VNI and IP/L3VNI
- Info is advertised in EVPN (control plane)
HOST DELETION & MOVE
Host Deletion: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.
Host Move: When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

VIRTUAL ROUTING AND FORWARDING (VRF)
- Supports independent policies per tenant
- Key for scaling and multi-customer separation
DISTRIBUTED ANYCAST GATEWAY (DAG)
- Same gateway IP+MAC on all VTEPs
- Enables local default gateway for hosts in MPLS network with IRB
- Supports mobility + optimal forwarding across BGP EVPN fabric

INTEGRATED ROUTING AND BRIDGING (IRB)
- Enables inter-VLAN routing inside EVPN
- Avoids central gateway → no "traffic tromboning"

SYMMETRIC IRB (L2 + L3)
- Routing/bridging on ingress + egress VTEPs
- Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)
- Scales well; clean separation of MAC and IP
- Requires L3 connectivity between all source/destination VTEPs for Type 2 routing (Done by configuring Type 5 routing)

ASYNMETRIC IRB (L2)
- Routing only on ingress, bridging on egress
- VXLAN uses destination VNI in both directions
- One L2 VNI per VLAN/Subnet
- Simple config, but requires all VLANs/VNIs on all VTEPs (scaling issues)

MULTIPROTOCOL BGP (MP-BGP)
Address Family Identifier (AFI): Category of information being carried
Subsequent AF (SAFI): Narrows down the specific type within that category
EVPN NLRI: Uses MP-BGP with specific AFI/SAFI. Supports multiple route types and attributes, unsupported routes are dropped by BGP

- Enables protocol-based VTEP discovery and host reachability via control-plane learning. Reduces flooding by replacing data-plane learning
- Extends BGP with multiprotocol capabilities (AFI/SAFI)
- Uses **MP_REACH_NLRI** and **MP_UNREACH_NLRI** for route advertisement and withdrawal (advertisises VPNv4 between PEs in MPLS)
- Uses the **VPNv4 NLRI** to distinguish between duplicate IPv4 prefixes

BGP CONTROL PLANE
- PEs learn MACs from local CES (data plane)
- MACs advertised via IRB (control plane)
- Uses Route Distinguisher for MPLS-based labels
- Remote PEs update L2 RIB/FIB with MAC and next-hop info
- Enables seamless L2 across IP/MPLS backbone

CDN
Origin Server: Central content source (original files), usually in a datacenter
Edge / CDN Server: Also called **Point of Presence (POP)**. Geographically distributed, caches content
Embedded PoP: Deployed directly inside ISP's local network
DNS Infrastructure: Directs users to optimal edge server (e.g. Geo-Routing)

KEY BENEFITS
Latency Reduction: Nearby edge servers reduce round-trip time
Availability: Failover and redundancy in case of node failure
Scalability: Handles traffic spikes via load balancing
Cost Optimization: Reduces backend and transit load on origin
DDoS Protection: Edge servers absorb attacks → not all traffic on one server
Global Load Reduction: Less long-distance traffic across the Internet

REQUEST ROUTING TECHNIQUES
- Decides which edge server should serve a client request
- Goal: Best performance (e.g. proximity, load, responsiveness)

DNS-BASED GEO-ROUTING
- Each edge has a unique IP
- DNS server picks closest/optimal edge server based on:
- Resolver IP location (not user!) → can cause wrong choice!
- GeoIP DBs (MaxMind, IP2Location), load, latency, business rules
- **DNS/O** and Client Subnet Extension (ECS)
- Resolver includes part of client IP in DNS request (e.g. /24 subnet)
- Authoritative DNS makes better decision based on actual client region
- Improves accuracy without revealing full IP, increases caching complexity

ANYCAST WITH BGP
- Same IP (e.g. 7.7.7.7) advertised from multiple locations
- BGP routing decides which path is "best" (AS-path, local pref, etc.)
- No DNS logic or per-client decision → pure BGP convergence
- **Pros:** Fast failover, simple, no app logic needed
- **Cons:** Less control, BGP ≠ best latency, route flapping risk

MANAGING TRAFFIC
Anycast: Modify BGP attributes/withdraw IP prefix. Not precise
Software: Shifting via Encapsulation tunnels (GRE)

LOAD BALANCING
Load balancing device: Efficient, reliable, High cost, single point of failure
ECMP: To router balances traffic. Simple, cheap, Adding/removing server changes hashing pop, re-hashing breaks TCP connections fix: GRE tunnel

HTTP CACHING & HEADERS
Caching is controlled via HTTP headers between clients, proxies, and servers
Cache-Control: Main directive (**no-cache, no-store, max-age, must-revalidate**, etc.)

Expires: Absolute expiration time (older method, replaced by **Cache-Control**)
ETag: Validator tag (version/hash), used with **If-None-Match**
Last-Modified: Timestamp used with **If-Modified-Since** for revalidation
Age: Time (in seconds) since response was fetched from origin
Validation: Client uses **Etag** or **Last-Modified**; server → 304 if unchanged

QOS
Internet is best effort: No guarantees, no QoS; all traffic treated equally (net neutrality), simple, scalable, but no delivery/order assurance or prioritization
Quality of Experience (QoE): Perceived service quality from user perspective
Route Pinning: Keeps flow on a fixed path to prevent oscillation (don't switch immediately to "better" path)
Routing-based: Path selection; QoS-aware routing; Choose best path
Node-based: Packet treatment; Congestion mgmt; queuing/scheduling/policing/shaping. "VNI happens to packets at each hop?"
PAK-Priority: System queue (routers) that prioritizes routing protocol packets
Class models: Realtime (VoIP > Video) > Control > Critical > Best effort

NETWORK PERFORMANCE METRICS
Bandwidth [Gbit/s]: Maximum transfer capacity of a link (width of pipe)
Throughput [Gbit/s]: Successfully delivered data (water that reached dest)
Latency / Delay [ms]: Time for packets to travel src → dest (length of pipe)
End-to-End Delay: Total time sender to receiver
One-Way Delay: From first bit sent to last bit received
Components: Transmission delay (time to push onto link), Processing delay (lookup, queuing), Propagation delay (physical travel time)
Jitter [ms]: Variation in delay between packets, caused by re-routing/queuing (Voip<30ms), Calc: no queue - queued delay. Doesn't impact TCP
Packet Loss Rate (PLR) [%]: Dropped packets due to congestion or errors

QUEUEING ALGORITHMS
First-In-First-Out (FIFO): Basic, no prioritization
Priority Queueing (PQ): Multiple queues, serve highest first; others may starve
Round-Robin (RR): One packet per queue in turn (fair, but ignores priority)
Weighted Fair Queueing (WFQ): RR with weights, n packets per queue
Class-Based WFQ (CBWFQ): WFQ with user-defined classes (logical queues), queue limits, max bandwidth guaranteed + max % of bandwidth
Low Latency Queueing (LLQ): Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol...)

QUEUE MANAGEMENT
Tail Drop: Drops packets when queue full; huge interruption of traffic → same as no connectivity
TCP Global Sync: Many TCP flows back off and restart simultaneously (AIMD) → link underutilization
TCP Starvation: TCP slows down after drops (congestion control), UDP doesn't → queues filled with UDP, TCP squeezed out
Random Early Detection (RED): Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their windows sizes
Weighted RED (WRED): RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic, max thresh → all packets dropped

POLICING VS. SHAPING
Policing (Inbound mostly): Drops packets that exceed configured rate limits
Shaping (Outbound): Buffers packets to smooth traffic bursts

QOS MODELS
BEST EFFORT
No guarantees, all traffic treated equally (follow Internet neutrality)
INTEGRATED SERVICES (INTSERV)
- End-to-end QoS (guaranteed for specific apps)
- Per-flow resource reservation
- Precise but not scalable (uses RSVP)
- App informs network of traffic traits, requests service before sending data
DIFFERENTIATED SERVICES (DIFFSERV)
- Scalable approach using marking. No hard guarantees, Cost-effective
- Divides traffic into classes based on business requirements
- Routers are configured with policy on how to prioritize classes

Marking (DSCP)
L3 Marking: ToS byte → DSCP (6 bits) + IP Precedence (3 bits) [overlapping]
- **DSCP** (6-bit in IP header) marks packets for QoS; classifies traffic
L2 Marking: Dot1q header → 802.1p/CoS bits
EXP: (3-bit in MPLS label) serves same purpose within MPLS networks; often mapped from DSCP.
Behavior (PHB)

Per-Hop behavior (PHB): Defines what routers actually do
Class Selector (CS): Compat with old IP Precedence, simple prio levels
Assured Forwarding (AF): Multiple classes + drop priorities
Expedited Forwarding (EF): Very high prio, e.g. VoIP

MODULAR QOS CLT (MQC)
Class Map: Define traffic classes (e.g., match voice or video)
Policy Map: Define actions for each class (e.g., limit, shape, priority)
Service Policy: Apply policies to interfaces or directions (in/out)

OTHER INFOS	
AD: Inter-protocol choice (e.g., OSPF vs RIP) → lower wins. Cost/Metric: Intra-protocol choice (e.g., OSPF path A vs B) → lower wins. Routing Preference Order (across protocols): - Most specific prefix - Lowest Administrative Distance - Static default route Administrative Distances: (Smallest Administrative Distance wins)	
Protocol	Distance
Connected	0
Static (Interface)	1
Static (Next Hop)	1
BGP External	20
EIGRP Internal	90
OSPF	110
ISIS	115
RIP v1/v2	120
EIGRP External	170
BGP Internal	200

EVPN BGP ROUTING TABLE INFOS
* = Would not be there if it was L2 VNI BGP Routing Table
Route Distinguisher: 172.16.255.101:32777
Route Type: 2
MAC Address Length: 48
MAC Address: 5254.00f8.29a8
***IP Address Length:** 32
***IP Address:** 10.10.0.100
***L3 VNI:** 50000
Remote VTEP IP Address: 172.16.254.101
L2 Route Target: 1:10
***L3 Route Target:** 65000:50000

leaf-03# show bgp l2vpn evpn 10.10.0.100
BGP routing table information for VRF default, address family L2VPN EVPN
Route Distinguisher: 172.16.255.101:32777
BGP routing table entry for [2]:[0]:[0]:[48]:[5254.00f8.29a8]:[32]:[10.10.0.100]/272, version 19897
Paths: (1 available, best #1)
Flags: (00000202) high32 00000000 on xmit-list, is not in l2rib/evpn, is not in HW
Advertised path-id 1
Path type: Internal, path is valid, is best path, no labeled nexthop
Imported to 2 destination(s)
AS-Path: NONE, path sourced internal to AS 172.16.254.101 (metric 81) from 172.16.255.1 (172.16.255.1)
Origin IGP, MED not set, localpref 100, weight 8
Received label 30010 50000
Extcommunity: RT:1:10 RT:65000:50000 ENCAP:8 Router MAC:5254.00ca.69ae
Originator: 172.16.255.101 Cluster List: 172.16.255.1

TODO:

```
<?xml:space="preserve">
<center>
  <route type=2*>
    <includegraphics[width=\linewidth]{route-type-2}>
    <route type=5*>
    <includegraphics[width=\linewidth]{route-type-5}>
  </center>
```

PRÜFUNG VORJAHR
NETWORK DESIGN

3-tier campus network: Default Gateway (D), QoS marking (A), STP Root Port (A), HSRP, VRRP or GLBP (D), "Simple" (C), OSPF Totally Stub Area (D), High availability (C)
Campus Design: used to reduce size of L2 domain: EVPN, MPLS

REST
MP_REACH_NLRI: Next hop, MAC Address
- VXLAN is a data plane technology which encapsulates Ethernet frames in UDP datagrams to tunnel layer 2 frames over a layer 3 network.
- The underlay network is unaware of VXLAN devices that connect to the physical switches are unaware of VXLAN.
- A route distinguisher is used to uniquely identify a route in combination with the destination prefix.

TODO'S
- p.74,75 - OSPF + IS-IS path cost calculation + path choosing - OSPF has to go through backbone
OSPF
- passive interfaces - flooding - O N1 / O N2 routes - route summarization - synchronizing the lsdb - Fast Reroute (FRR)
IS-IS
- ES-IS, IS-IS: details
BGP
- next-hop-self - comparison to IGP - Multihop sessions - NLRI - Aggregate impact (TE) - rework - Best path calculation - Idle → Connect → Active → OpenSent → OpenConfirm → Established - MED explanation - default external cost = 20?
AVAILABILITY
- everything: thanks, past me, that was very informative
MULTICAST
- Link-local multicast (48) - IPv4 MAC Address mapping - IGMP and the querier/Role in Dense Mode/Challenges
TOPOLOGY
- p.59-61 - Slides 40-50 - pros/cons of different topologies
MPLS
- VPWS, VPLS (slides MPLS_VPN, 50,51)
EVPN
- Route types - IRB diagram
QoS
- Types of traffic - Queuing input/output buffer - IntServ vs DiffServ - QoS classification - DSCP,PHB,CS,AF,EF - rest - Node based vs Routing based?
TODO:
- github repos for IS-IS and OSPF labs - why ibgp needed for ebgp - ip address otherwise might not be known in AS internally - cdn lab (tags?) - qos: why do we need it? - how does it work conceptually - voice QoS sensitive to? - spick lukas QoS+ - cisco config: bgp (slides VXLANEVPN 34), mpls, (ospf)
TODO:
notes 37+